History

This methodology was originally developed by Michael Stone in the fall of 2017. It was adopted with minor amendments (can be identified by red font) to facilitate a 2019 research project titled "Ranchland Dynamics 2.0." For the Ranchland Dynamics 2.0 research project, the GIS work and owner homogenization were completed by Jackson Rose under the direction of Dr. Julia Haggerty. Substantial revisions and additions were added by Michael Stone and finalized on [insert date] for inclusion in his professional paper. This version assumes the user has basic competency with Microsoft Excel, SQL, and basic knowledge of geographic information science and the QGIS program.

About the Cadastral

The Montana Cadastral Framework shows the taxable parcels and tax-exempt parcels for most of Montana. The parcels contain selected information such as owner names, property and owner addresses, assessed value, agricultural use, and tax district information that were copied from the Montana Department of Revenue's (MTDOR) ORION tax appraisal database. The data are maintained by the MTDOR, except for Ravalli, Silver Bow, Cascade, Missoula, Lake, Flathead and Yellowstone counties that are maintained by the individual counties. The Revenue and county data are integrated by Montana State Library staff. Each parcel contains an attribute called ParcelID (aka geocode) that is the parcel identifier. Large areas of public land in Flathead County are missing from the data. The parcel boundaries were aligned to fit with the Bureau of Land Management Geographic Coordinate Database (GCDB) of public land survey coordinates. Parcels whose legal descriptions consisted of aliquot parts of the Public Land Survey System (PLSS) were created from the GCDB coordinates by selecting and, when necessary, subdividing public land survey entities. Other parcels were digitized from paper maps and the data from each map were transformed to fit with the appropriate GCDB boundaries.

Geospatial Data Acquisition

1. To perform a bulk download of large files it is best to use a file transfer protocol (FTP) application. Filezilla Client (clicking the hyperlink will take you to the download) is a free and widely used program that can be downloaded from https://filezilla-project.org/.

2. To acquire historical cadastral data in vector format[1], it is no longer necessary to submit a data request to the Montana State Library (MSL) because there is now a dedicated archival folder on their FTP server called "CadastralHistoric." In the FTP client, use the following host address: ftp.geoinfo.msl.mt.gov/Data/Spatial/MSDI/Cadastral/CadastralHistoric/ to access the file directory.

3. Once inside the CadastralHistoric directory there are several zipped files. All parcel layers have the following naming convention: CadastralYYYYMMDD. Each of these files is the statewide

[1] The Montana State Library provides cadastral data in a proprietary vector file developed and regulated by ESRI called a "shapefile" and it has the ".shp" file extension.

parcel shapefile published for the indicated year, month, and day. So, if you were interested in the parcel composition of Montana for the year of 2013, you have two options

4. correspond with bi-annual property taxation. To download the files simply select, drag, and drop the desired items within a directory on your local file system. Once unzipped, the file will consist of the relevant shapefiles and the metadata. It's important to note that the full CAMA databases are not included in this archive.

5. To acquire the current cadastral, use the following host address in your FTP client: ftp.geoinfo.msl.mt.gov/Data/Spatial/MSDI/Cadastral/Parcels/. This will take you inside the directory that contains the most recent parcel layers for each county and the whole state. Each county is indicated by name and the statewide cadastral is aptly titled "Statewide." Each county folder consists of a geodatabase and a folder containing the parcel shapefile of the geography. The statewide folder has two more items: an update map which provides relevant metadata and the CAMA database produced by the Montana DOR. To download the files simply select, drag, and drop the desired items within a directory on your local file system. Within the current Montana Cadastral Framework (MCF) parent directory, there is a subdirectory 'Parcels' that contains current county parcel datasets separated by county. This dataset is updated multiple times per year, but no historical county parcel datasets are readily available within the MCF. Acquiring historical county parcel data requires calling the Montana State Library (MSL) at 406-444-5354 (this number was obtained from the metadata associated with the MCF). At this point, the MSL will provide temporary access to historical data through their File Transfer Protocol (FTP) site, where zipped folders containing statewide parcel data for requested years are available. While it is possible that MSL can select out specific counties upon request, this has not been verified. If historical parcel data is not needed, skip the "Seperating out specific counties" section, and move on to the "Creating a unique owner code" section of this methodology.

County Export

For the historical cadastral data, sub-state geographies must be manually exported from the provided statewide shapefile. Here, instructions are provided for extracting individual counties, but an extension of these concepts will allow for custom geographies as needed. The most current version of the cadastral should have individual counties separated and available for download (refer to step 4 in the Geospatial Data Acquisition section).

1. Download the requested statewide parcel dataset from the MSL FTP site. Acquire the desired annual statewide cadastral shapefile (see Geospatial Data Acquisition section for instructions).

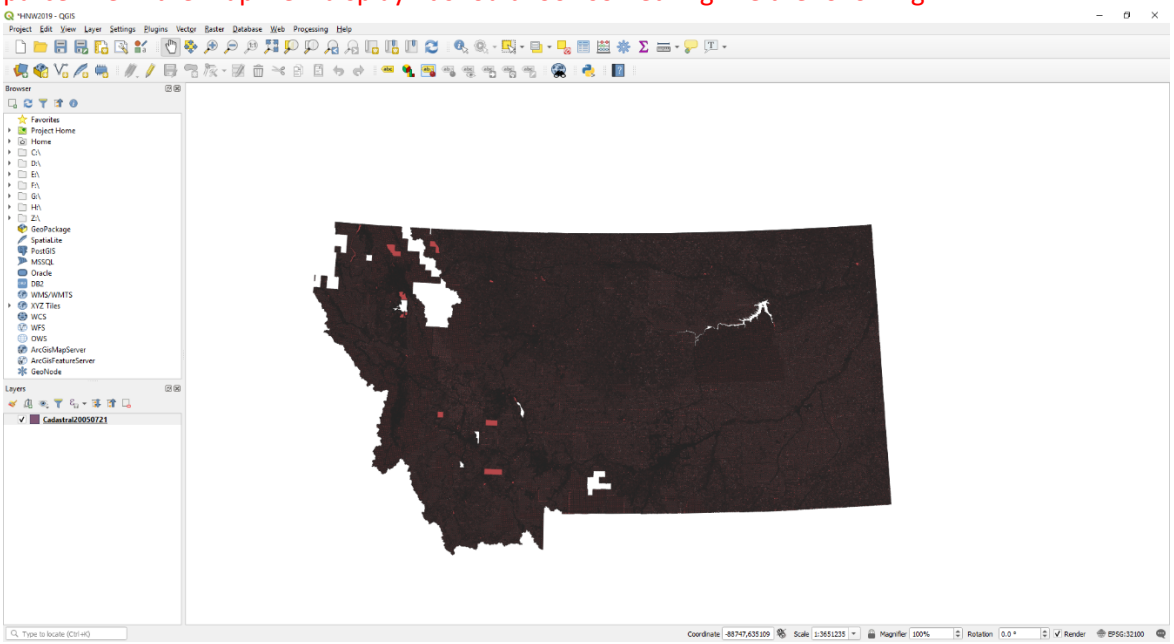
2. Move the downloaded file to a project workspace and extract the files contained in the zipped folder.

3. Read the attached metadata document, making sure to note counties that have data missing.

4. Open QGIS.

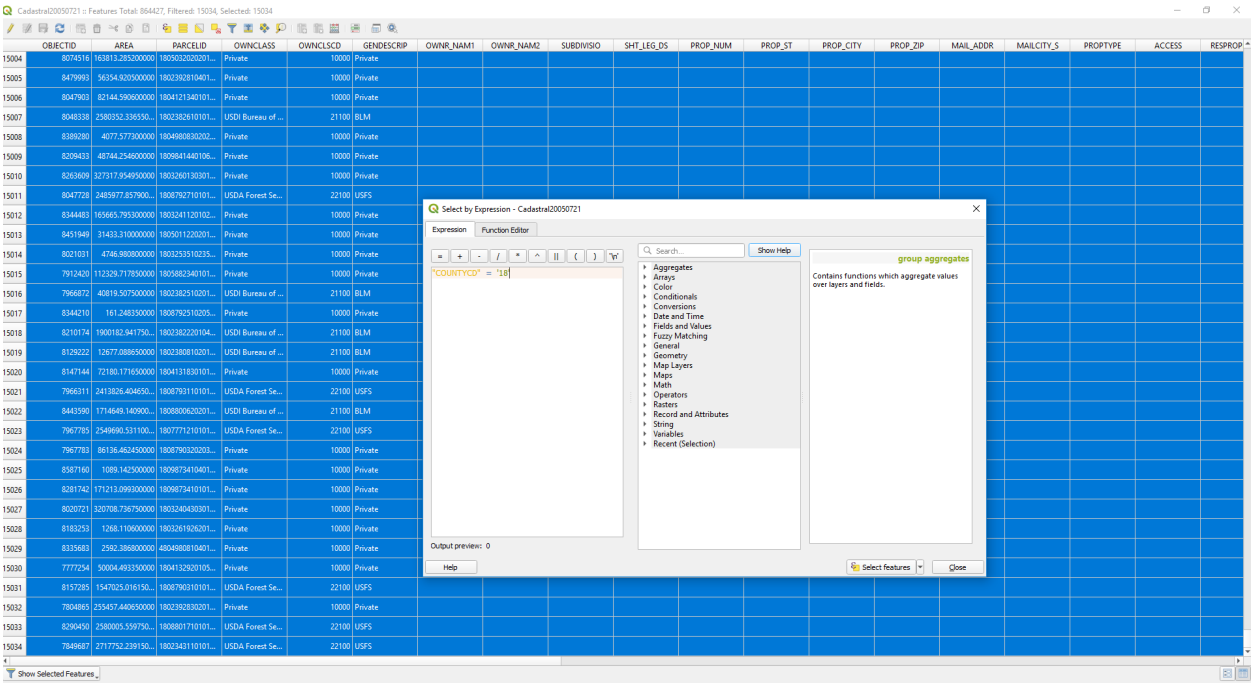
Date: 3/4/2020

5. Import the shapefile (that is, the .shp parcel file previously unzipped). You should then see the parcel file in the map view display. It should look something like the following:



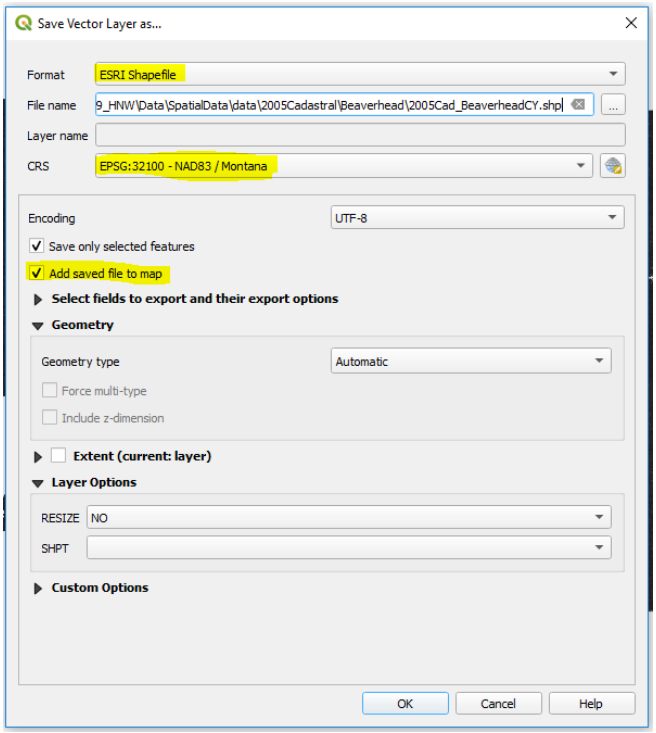
6. From this statewide parcel file, we want to select out a specific county to consolidate the ownership information. For this demonstration, the Beaverhead Cascade County dataset will be selected. Open the attribute table for the statewide vector file. In the attribute table toolbar, select the button to “Select by expression”. This will open the expression builder window. Use the following expression to select parcels in Cascade County: "COUNTYCD" = '18'2. Click on “Select Features” in the bottom right part of the Select by Expression window. In the attribute window click on the drop down menu in the bottom left part of the window and select “Show Selected Features”. The results should mirror the following:

Date: 3/4/2020



7. We now want to export and save the Cascade County parcels as a separate shapefile. Make sure you have a folder created in your workspace to place this extracted dataset that's separate from the source data folder. This keeps the data source as is and allows future use without concern for modification of the data. Beaverhead

8. Right click on the Cadastral vector file in the "Layers" window, and select "Export", "Save Selected Features As". Set the parameters to the following with an appropriate file name:



9. The new file should be visible in the "Layers Panel" window in QGIS.

ORION Database Acquisition

The attribute table associated with the cadastral shapefile is derived from the ORION database and does not contain all the data associated with properties in the state. We will need some of the additional information provided in the database to supplement the data provided in the attribute table. To retrieve this data, the user will need to have an SQL database client installed on their system to open and query the relevant ORION database files. A free option is SQL Server Express and Microsoft SQL Server Management Studio. It is beyond the scope of these instructions to provide guidance on installing these programs. Help is available through Microsoft for installation support as well as guidance on connecting to a database.

1. Use your FTP client to access the Montana State Library FTP server at this address: ftp://ftp.geoinfo.msl.mt.gov/Data/Spatial/MSDI/Cadastral/ORION_SQLDatabases.

2. Once inside the ORION_SQLDatabases directory, you should see several zip files, a change log, and two PDF documents. If you already know the county number for the county or counties you are interested in you can simply drag and drop those zip files into your designated directory for SQL databases. If you do not know the relevant county code(s) then you can reference the numbers in either of the CountyName" or "CountyNumber" PDF documents in the directory. If you want the entire state, download the "STATE-WIDE" zip file.

3. In addition to the geographies of interest, you should download the "BEGIN HERE" zip file which includes important metadata on the database files and structure.

4. Once you have unzipped the database files in your chosen directory, you will need to connect to the database file through an SQL database client such as Microsoft SQL Server Management Studio (MSSMS).

5. Once connected to the ORION database of interest, there are two queries that can be used to produce two tables for export. The first table of interest will consist of deed data associated with properties in the geography of interest. The query below was used to pull all recorded deed records between January 1st, 2005 and December 31st, 2019.

```
SELECT * FROM dbo.Deed d
WHERE d.DeedDate BETWEEN '2005-01-01' AND '2019-12-31'
ORDER BY PropertyID;
```

1. The second table of interest provides detailed ownership information for the property within the geography for the 2020 tax year. The query below uses the relationship between the Owner, PartyName, and PartyAddr tables to pull a subset of records for the established criteria of the query:

```
SELECT p.GeoCodeSearch, p.PropertyID, p.Township, p.Range, p.Section,
p.SubSection, p.SubBlock, p.SubLot, p.LegalDescription,
p.CertificateOfSurvey, p.COSParcel,
p.Subdivision, p.PropType, p.Addr_Number, p.Addr_PreDirectional,
p.Addr_Street, p.Addr_RoadSuffix, p.Addr_PostDirectional, p.Addr_City,
p.Addr_State, p.Levy_District,
p.TimestampChange, o.TaxYear, o.PartyID, o.PercentOwnership,
o.PrimaryOwner, o.InterestType, o.TimestampChange, n.PartyNameID,
n.DefaultName, n.FullName, n.NameType,
n.NameType_Desc, a.PartyAddrID, a.DefaultAddress, a.Address1, a.Address2,
a.Address3, a.City, a.State, a.ZIP, a.Country, d.Book, d.Page, d.DeedDate,
d.DocNumber,
d.DocType, d.DocType_Desc, d.RecordedDate
FROM [Property] p
JOIN dbo.Owner o ON p.PropertyID = o.PropertyID
JOIN dbo.PartyName n ON o.PartyID = n.PartyID
JOIN dbo.PartyAddr a ON n.PartyID = a.PartyID
JOIN dbo.Deed d ON p.PropertyID = d.PropertyID
WHERE p.TaxYear = 2020 AND o.TaxYear = 2020 AND p.PropType IS NOT NULL
ORDER BY o.PropertyID;
```

2. Export both the resulting table to a comma separated values (.csv) file format. They are now ready to be imported into Microsoft Excel.

Creating a unique owner code[2]

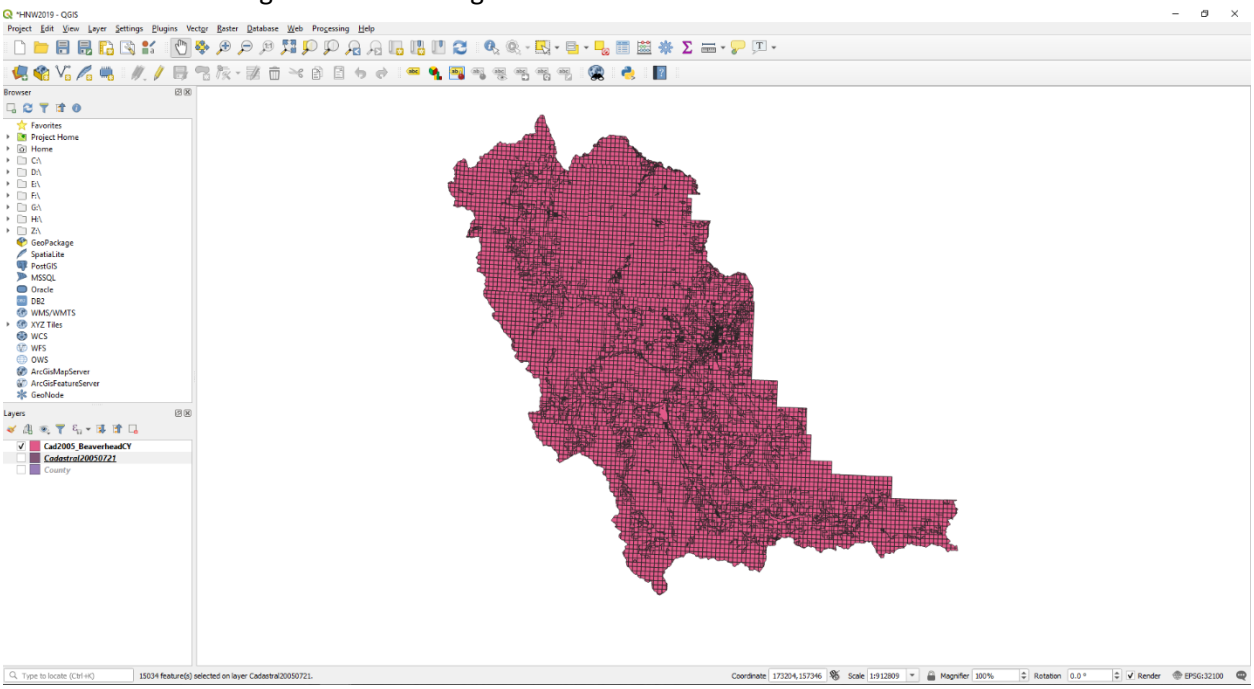
1. Download the desired county parcel dataset from the Montana Cadastral (see Data Acquisition and County Export, above, for instructions).

[2] The following steps will involve using several formulas included in Microsoft Excel. These formulas operate like functions that take arguments with optional parameters. Given the variable nature of the arguments, I will use the following conventions to indicate cell locations in a variable format. The Greek alphabet will be used to indicate a variable column letter. For variable row numbers, italicized *x*, *y*, and *z* will be used. So, if the relevant cell for a formula is A492, then this will be represented as a variable "α*x*" in the formula. This allows the formulas used in the process to be provided without referring to specific cells used in the process that may be different between datasets or user made changes that offset from these instructions.

2. Move the downloaded zip file to an appropriate project workspace and unzip the file. Make sure to create a new directory to contain the files associated with the shapefile as there will be multiple individual files necessary for the shapefile to work.

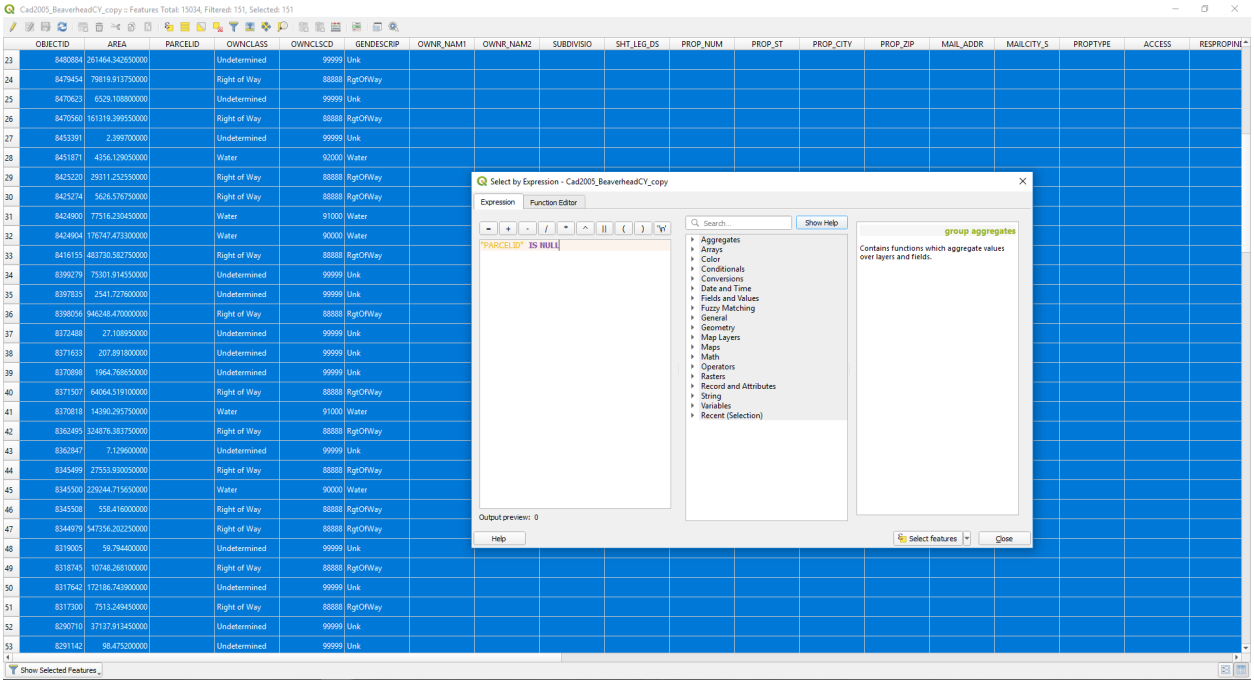
3. Open QGIS.

4. Add the vector file to your project. You should then see the parcel file in the map view display. It should look something like the following:



10. The parcel file, as is, likely has numerous null PARCELID cell values.[3] We want to keep a record of these so we can fully account for the data but we also want to remove these before exporting the attribute table to a .csv file for analysis. To achieve this separation, we will be creating two new shapefiles from the original. One will consist of all the null PARCELID features and the other will consist of non-null PARCELID features. Open the attribute table for the county shapefile. In the attribute table toolbar, select the button to "Select by expression". This will open the expression builder window. Use the following expression to select all null PARCELID features: "PARCELID" IS NULL. Click on "Select Features" in the bottom right part of the prompt window. In the attribute window click on the drop-down menu in the bottom left part of the window and select "Show Selected Features". The results should mirror the following:

[3] These polygons largely consist of public rights-of-way and waterways but some are unidentified properties or some other anomaly.

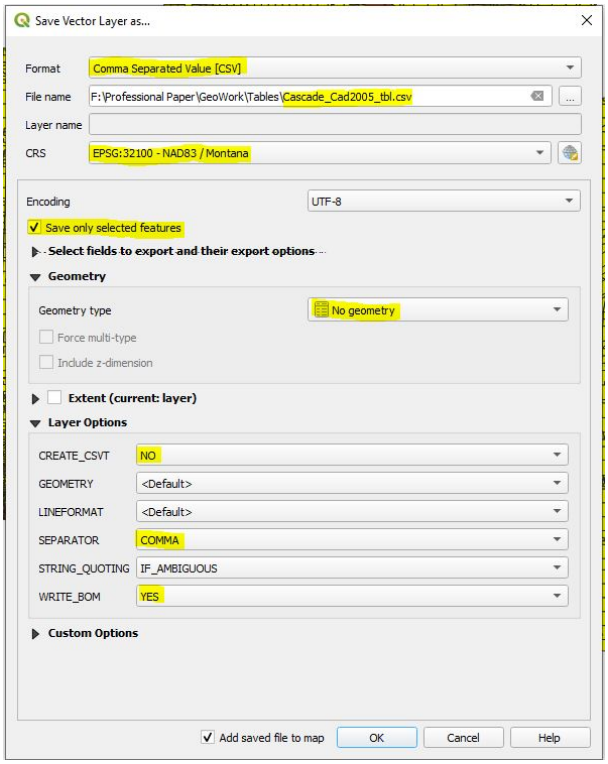


5. Export the selection of features to a shapefile. Make sure you provide a name for the file that identifies it as the null PARCELID shapefile when you save it in your designated workspace folder.

6. With the attribute table and expression builder still open for the original county shapefile, we now want to create an inverse expression to select all non-null PARCELID features. Use the following expression: "PARCELID" IS NOT NULL. Click on “Select Features” in the bottom right part of the prompt window. In the attribute window click on the drop-down menu in the bottom left part of the window and select “Show Selected Features”. Repeat step 8, above, for this selection of features. Make sure you provide a name for the file that identifies it as the PARCELID shapefile when you save it in your designated workspace folder.

7. Now we are ready to export the attribute table with all non-null PARCELIDs associated with the county parcel geography. We already have the desired selection of features from the last step so we can right-click the vector file in the “Layers” window. Select the option to “Export”, "Save Features As". Set the parameters to those displayed in the picture below with the file name set

to an appropriate name for your chosen county:



Once the table has been exported you can close out of QGIS. Open a new worksheet in Microsoft Excel.

Select the “Data” tab in the toolbar and click on the option to import “From Text/CSV.” A prompt will appear. Navigate to the .csv file you exported from QGIS and click “Import.”

8. The import prompt will appear with a preview of the .csv table in excel spreadsheet format. This will open the table in the worksheet with a default table formatting applied. Make sure the PARCELID is treated as a text data type rather than numerical (this may require transforming the data in the Excel import prompt). Complete the import and the spreadsheet should open. Name this sheet “Original.” This sheet will be left alone to preserve the original data source. Copy the “Original” sheet into a new sheet within the same book. Label this sheet “Mod1”.The data in the Mod1 sheet will be the base reference for the assignment of unique owner, place, and address identifiers for the entire original dataset.

9. Copy and paste the field from the attribute table that contains the name of the property owner(s) into a new worksheet.OWNR_NAM1 an empty Run the Remove Duplicates tool to remove duplicates.[4] If there is only one owner name field, name the worksheet “OwnerCodes” and skip the next step.

[4] The formatting of this field varies depending on the dataset, for the 2005 cadastral there are two relevant fields “OWNR_NAM1” and “OWNR_NAM2” whereas more recent datasets only have one field titled “OwnerName.” If there are two fields, make sure both fields are used. If only one field is selected it will result in a mismatch. For more recent datasets it is a single field titled “OwnerName.”

10. If the dataset has two owner name fields, use the CONCAT formula to concatenate the fields containing the owner names separating them with a space. Copy and paste the formula for all the owners. Name the worksheet “OwnerCodes.”

11. For property owner names we will use a nine-digit code reserving “000000000” for later use. To create a manual unique identifier, select the first cell to the right of the first property owner name. Type a “1” in that cell and press “ENTER.” You should now be in the cell directly below it. In this cell type “=αx +1” where αx is the cell with the value “1” in it. This will allow us to recursively generate a number for each row by simply copying the “=αx +1” formula and pasting it in all the subsequent cells until the last property owner name is reached. Some owner name cells will be empty and it’s important to make sure these are captured in the process since other information may be provided for the feature. Return to the top and click on the first cell to the right of the cell with the value “1.” Use the TEXT formula to convert the numbers we just generated into the text data type and provide the format as a nine-digit number “000000000.” Copy and paste the formula for all the property owners (each row). Title the field “OwnCode.” The results should look something like this:

Concat	OwnCode
CABOOSE MINING COMPANY	000000001
MASHBURN SHARRON LEE ETAL HALL PAAVO KURT	000000002
MCDONALD NORMA S	000000003
FRITZ ROBERT A	000000004
TOWN OF NEIHART	000000005
JUNTUNEN ANTHONY E & SHERRY L	000000006
DAVIS JOHN H	000000007
GOINGS DEAN V	000000008
NEIHART CITY OF	000000009
SUPER JAMES A	000000010
MASTERS GLENNA J & RAY	000000011
JOHNSON DARVIN G	000000012
CROFF CHRISTIE J ETAL SCHWARTZ SANDRA L	000000013
GREENE JOHN WILLIAM JR	000000014
FLY FISHERS INN INC	000000015
CANYON CONSTRUCTION OF MONTANA	000000016
DEARBORN HOLDING COMPANY	000000017
MISSOURI RIVERSIDE INC	000000018
UNITED STATES OF AMERICA	000000019
FILLINGER STEVE & RHONDA	000000020
OBRESLEY JOSEPH N ESTATE	000000021
ENLOE HUGH D & JANET M	000000022
CUBS DEN LLC	000000023
PRUITT JAMES E & JANE	000000024
BURLINGTON NORTHERN SANTA FE R	000000025
SMITH RIVER REC ASSOC	000000026
SCHOOL DISTRICT #84	000000027
GERHART JOHN V	000000028
EVENING POST PUBLISHING COMPAN	000000029
NORTHWESTERN CORPORATION	000000030
URSULINE CONVENT OF THE HOLY F	000000031
LATTER DAY SAINTS CORPORATION	000000032
QUIET DAY MANOR PARTNERSHIP	000000033
THOMPSON EUGENE R & CHARLENE R	000000034
MEYER JOHN WILLIAM	000000035
CREVELING JACK R & DOROTHEA M	000000036

12. Use “Format as Table” in the toolbar to format the owner code data as a table. Select a format and make sure the table headers are included. We will use this table to import the owner codes into the Mod1 table.

13. Format the Mod1 table with the “Format as Table” button if it is not already a table. Click on the first cell that’s outside the table and type “OwnCode.” If the owner name consisted of two or more fields, use the CONCAT formula to concatenate the owner names from OWNER_NAM1

and OWNR_NAM2, separating them with a space so that the concatenated owner name matches the results from the OwnerCodes sheet. This field will allow us to import the codes from the OwnerCodes table.

14. In the Mod1 OwnCode field, use the VLOOKUP formula to import the matching OwnCode for the respective owner name making sure that the FALSE argument is chosen for the formula to ensure an exact match. Since the dataset in Mod1 is formatted as a table, it will automatically populate the rest of the entries.

15. Create a new worksheet and name it “OwnAddr”. In this sheet we will be assigning a unique identifier to unique addresses. The steps here parallel the previous steps for property owner names. Locate the fields containing the property owner mailing address in Mod1. Depending on the year, these field names will vary, for the 2005 data there are two fields titled “To start, copy

Address	AddrID
1029 CHOTEAU ST HELENA, MT 59601	0000001
1845 ARBOL VERDE WAY LAS VEGAS, NV 89119	0000002
1035 ALPINE HEIGHTS RD ALPINE, CA 91901	0000003
PO BOX 91 NEIHART, MT 59465	0000004
PO BOX 36 NEIHART, MT 59465	0000005
PO BOX 112 NEIHART, MT 59465	0000006
PO BOX 610634 BIRMINGHAM, AL 35261	0000007
409 N MAIN ST NEIHART, MT 59465	0000008
514 7TH ST S GREAT FALLS, MT 59405	0000009
617 5TH AVE N GREAT FALLS, MT 59401	0000010
PO BOX 933 MONARCH, MT 59463	0000011
PO BOX 126 MONARCH, MT 59463	0000012
2706 FAIRMONT RD ANACONDA, MT 59711	0000013
2629 OLD US HIGHWAY 91 CASCADE, MT 59421	0000014
104 BALD EAGLE DR CASCADE, MT 59421	0000015
4 COOPER DR CASCADE, MT 59421	0000016
3103 OLD US HIGHWAY 91 CASCADE, MT 59421	0000017
228 75TH ST N GREAT FALLS, MT 59402	0000018
412 CALVERT RD STOCKETT, MT 59480	0000019
5382 US HIGHWAY 89 BELT, MT 59412	0000020
PO BOX 871 MONARCH, MT 59463	0000021
PO BOX 70 MONARCH, MT 59463	0000022
PO BOX 24 MONARCH, MT 59463	0000023
PO BOX 961089 FORT WORTH, TX 76161	0000024
1699 MILLEGAN RD GREAT FALLS, MT 59405	0000025
1508 MILLEGAN RD GREAT FALLS, MT 59405	0000026
6203 US HIGHWAY 89 BELT, MT 59412	0000027
134 COLUMBUS ST CHARLESTON, SC 29403	0000028
125 S DAKOTA AVE SIOUX FALLS, SD 57104	0000029
GENERAL DELIVERY ULM, MT 59485	0000030
50 E NORTH TEMPLE SALT LAKE CTY, UT 84150	0000031
PO BOX 1643 GREAT FALLS, MT 59403	0000032
PO BOX 69 CASCADE, MT 59421	0000033
PO BOX 645 CASCADE, MT 59421	0000034
PO BOX 373 CASCADE, MT 59421	0000035
PO BOX 113 CASCADE, MT 59421	0000036

columns D (MAIL_ADDR”) and “ and E (MAILCITY_S” that contain the street address, city, state, and zip code. For the 2020 dataset, there are three address fields (number in sequence) with additional fields for city, state, and zip code. Create a new field in the Mod1 worksheet titled “OwnerAddr.” Use the CONCAT formula to concatenate all the address fields with a comma between each field. Copy the OwnerAddr field into the OwnAddr worksheet and use the Remove Duplicates tool to remove all the duplicate addresses. Add a new field called “AddrID” and generate a seven-digit code following the same procedure as described above for the owner names. Format the data as a table.

16. Now we need to import the address IDs into the Mod1 worksheet. Create a new field called “AddrID” next to the “OwnCode” field in Mod1. Use the concatenated addresses from the OwnerAddr field in Mod1 and the VLOOKUP formula to import the address IDs from the OwnAddr worksheet.) from our Mod2 sheet to our new OwnAddr sheet. Keep in mind that the columns copied depend on the separation of address data in our Mod2 sheet. For example, if all the address data (street address, city, and state) were combined in one column, we could skip the concatenate step. . In column C of the OwnAddr sheet (with columns A and B being the MAIL_ADDR and MAILCITY_S values respectively), enter in a formula that looks something like

this “=CONCAT(A2, ", ", B2)" and then fill the formula to the end of the range. This should result in a combination of columns A and B into column C that looks like this:

17. Create a new worksheet titled “Mod2.” Copy all the data contained in Mod1 and “paste as values” in the Mod2 sheet and format it as a table. This preserves the formulas used to import unique identifiers in Mod1 for data checking. Sort the owner name field alphabetically from A to Z. Now we need to import the address codes (“AddrCode”) from our OwnAddr sheet into the Mod2 table just as we did with the OwnCodes. First, let's create a temporary concatenated "OwnerAddr" column in the Mod2 table. Copy the CONCAT formula above (although note you will have to change the columns) into a column immediately adjacent to the "MAILCITY_S" column and fill to the bottom of our values (again remembering to delete values of "," at the bottom of the range). Next, add a column "AddrCode" in the first empty column of the Mod2 sheet (should be to the right of the "OwnCode" column). We will use a similar VLOOKUP function to fill in this "AddrCode" column, in this case it looks like: =VLOOKUP(F2, OwnAddr!A:B, 2, FALSE). This formula should fill to the bottom of the sheet. Once the AddrCode values are in the Mod2 sheet, copy and paste the values (we need to replace the formula results) in column H back into column H and delete the concatenated entries column (column F in this case). Sort the addresses alphabetically (A to Z) for the "MAIL_ADDR" field.

18. Create three new columns in the Mod2 table. Name the first new column “OwnCodeMatch.” Name the second column “AddrIDMatch.” Name the third column “CombMatch”. In "OwnCodeMatch" enter into the first cell under the field name the following formula: “=IF(OR(α*F*21 = α2 F1, α2 F2= α3 F3),TRUE)” where the α column is the column with the unique owner codes. This formula looks for similarity between the cell above or below the current cell and, if it finds a match, enters “TRUE” into the cell, otherwise, it will produce “FALSE” or an error code. Similarly, we will make a formula that does the same thing with unique address codes in the AddrIDMatch field. In the CombMatch field, we want to create a statement that keeps the AddrID value under two conditions: if the value of OwnCodeMatch and AddrIDMatch are both false or, if AddrIDMatch is true. This is the formula "=IF(OR(AND(α_1H2=FALSE, α_2I2=FALSE), α_2I2=TRUE), β2G2,FALSE),"." where α_1 is the OwnCodeMatch field, α_2 is the AddrIDMatch field, and β is the AddrID column. The CombMatch field helps us identify the errors in addresses without having to manually review each entry. Where there is a “FALSE” value in the CombMatch field, we know that this means that OwnCodeMatch is TRUE but AddrIDMatch is FALSE. This identifies matching owner codes that have a difference in the address. This is very useful because the owner address provides a quick way to get through all the variations in the way that owner names are reported. The example in the image below provides an example from the Beaverhead 2005 Cascade County cadastral dataset.

000002270	WELLS FARGO BANK NA	SC	29715	FORT MILL, SC	4509126890	3476 STATEVIEW BLVD FORT MILL, SC 29715	0002001	TRUE	TRUE	TRUE	0002001
000002270	WELLS FARGO BANK NA	SC	29715	FORT MILL, SC	4509126890	3476 STATEVIEW BLVD FORT MILL, SC 29715	0002001	TRUE	TRUE	TRUE	0002001
000002270	WELLS FARGO BANK NA	SC	29715	FORT MILL, SC	4509126890	3476 STATEVIEW BLVD FORT MILL, SC 29715	0002001	TRUE	TRUE	TRUE	0002001
000002270	WELLS FARGO BANK NA	SC	29715	FORT MILL, SC	4509126890	3476 STATEVIEW BLVD FORT MILL, SC 29715	0002001	TRUE	TRUE	TRUE	0002001
000002270	WELLS FARGO BANK NA	MD	21703	FREDERICK, MD	2402130325	7495 NEW HORIZON WAY FREDERICK, MD 21703	0017220	TRUE	FALSE	FALSE	FALSE
000002270	WELLS FARGO BANK NA	SC	29715	FORT MILL, SC	4509126890	3476 STATEVIEW BLVD FORT MILL, SC 29715	0002001	TRUE	FALSE	FALSE	FALSE
000017892	WELLS FARGO FINANCIAL MONTANA	IA	50323	URBANDALE, IA	1904979950	4143 121ST ST URBANDALE, IA 50323	0015789	FALSE	FALSE	FALSE	0015789
000018636	WELLS JAMES R	MT	59405	GREAT FALLS, MT	3001332800	623 22ND ST S GREAT FALLS, MT 59405	0016464	TRUE	TRUE	TRUE	0016464

19. Create a new sheet and name it “OwnAddrCheck”. Copy the OwnCode, CombMatch, OwnerName, and OwnerAddr fields from "OWNR_NAM1", "OWNR_NAM2", "MAIL_ADDR""MAILCITY_S" Mod2 (the OwnerAddr field should be the concatenated address field used to lookup the address ID) and paste as values in the OwnAddrCheck worksheet. OWNR_NAM1, OWNR_NAM2, MAIL_ADDR"MAILCITY_S

20. At this point, we will begin resolving mismatches indicated by “FALSE” values in the CombMatch field. Resolving the "FALSE" values in the CombMatch field must be done manually because it relies on making judgements about why the field is FALSE. Highlight the entire column and use the Find and Replace tool to locate the “FALSE” values in the column.

0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0514	AMERICAN COLLOID CO	ATTN TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0516	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN EST	IL	60192-3702
0516	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN EST	IL	60192-3702
0516	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN EST	IL	60192-3702
FALSE	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0516	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN EST	IL	60192-3702
0516	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN EST	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702
0515	AMERICAN COLLOID CO	ATTN. TAX DEPT	2870 FORBS AVE	HOFFMAN ESTATES	IL	60192-3702

The image below above displays the first FALSE value for the 2017 Phillips County cadastral dataset. Notice that there are multiple data entry inconsistencies that are producing mismatches that should be have the same address ID. Also, notice that slight differences in the entry of addresses have created multiple address codes for what is essentially the same address. Data entry errors, such as this example, are easy to resolve by simply replacing the “FALSE” cell value with a consistent address ID. Another common issue is cases where the owner name is the same but paired with different addresses. These cases require considering the character of the owner name and its relation with the address. When the owner is a company, the same owner name under a different address can be assumed to be the same entity due to the regulatory

framework around business names which must be registered with the Montana Secretary of State's office. With private individuals, we cannot assume that the these are the same person without further evidence. The table below provides a list of common issues that led to a FALSE result in the CombMatch field for the 2020 Cascade County cadastral. Each issue has a corrective step provided.

Issue	Example	Corrective	Verification
Data entry	Inconsistent data entry for identical information, spelling errors, etc.	Replace "FALSE" cell value with the relevant AddrID value	
Matching owner name with multiple owner addresses (company)	Business provided more than one mailing address to the MT Department of Revenue.	Replace "FALSE" cell value with the relevant AddrID value for the unique address provided in the AddrIDSup field	Property Party Query from ORION Database
Matching owner name with more than one owner and multiple owner addresses (non-company)	Married couple that provided more than one mailing address to the MT Department of Revenue.	Replace "FALSE" cell value with the relevant AddrID value for the unique address provided in the AddrIDSup field	Property Party Query from ORION Database
Matching owner name with multiple owner addresses (trusts)	Trust provided more than one mailing address to the MT Department of Revenue.	Replace "FALSE" cell value with the relevant AddrID value for the unique address provided in the AddrIDSup field	Property Party Query from ORION Database
Individual owner name with multiple owner addresses	The ambiguity may due to the same owner with different addresses but may be two different people with the same name.	Seek further evidence from ORION Database or Deed documents.	Property Party Query from ORION Database and Deed Query

21. Resolving the "FALSE" entries mostly consists of finding what created the error and, often, simply assigning the correct code to the respective cell. For example, with the AMERICAN COLLOID CO in the image above, "0516" is already assigned to that owner. Correcting it consists of placing that value in the cell. In other cases, there may not be a code assigned because all unique entries for that owner are FALSE. These cases are trickier, and we have to consider the possibility that there are multiple owners with identical names (in other words we can not necessarily assume that it is the same owner with different addresses unless that is obvious in the "OWNR_NAM1" value). If we cannot determine they are common owners a unique address code must be assigned, we need to make sure that we assign an unused code. A simple way of doing this is to copy and paste the current list of used codes in a new sheet, remove the duplicate values, and find an unused code value. Alternatively, if "1891" is the highest code

value for our four-digit address code, then you can start at 1892 and work up from there, making sure you keep track of used codes as you go. The best way to ensure you keep track is to add every additional code used at the bottom of the OwnAddr sheet. For Beaverhead County, there were 42 FALSE values.

22. As progress is made in correcting "FALSE" entries it becomes apparent that there are other issues that need to be accounted for. One being the issue pointed out in step 32 with slight differences in address data entry. Another issue is created by using addresses to assign owner codes with cases where the address is the same for different owners. This is commonly the case with General Delivery addresses from town post offices. To make this process quicker, we will import the Owner Codes from the OwnerCodes sheet into the OwnAddrCheck sheet using the VLOOKUP function. The VLOOKUP function here is similar to the first time we used it: =VLOOKUP(C2, OwnerCodes!A:B, 2, FALSE). Create the OwnCode column to the right of the CombMatch column and populate it with the VLOOKUP function (remember that values will be "0" until you scroll to the beginning of OWNR_NAM1 values that are not blank).

23. Create four columns with the following titles: AddrUMatch, OwnCodeUMatch, AddrDMatch, OwnCodeDMatch. Each of these columns will consist of an IF function that will produce a "TRUE" value if the given cell matches with the cell above or below (Note: make sure the CombMatch column is sorted A to Z to order on addresses). Columns with a "U" in the name look up, if there is a "D" it looks down for the CombMatch column and OwnCode column. Here's an example formula: "=IF(A2=A1,TRUE,FALSE)". The values in these columns are informative when interpreted through a matrix. For example:

AddrUMatch	OwnCodeUMatch	AddrDMatch	OwnCodeDMatch	Description
TRUE	TRUE	TRUE	TRUE	Name and address above and below match.
FALSE	FALSE	FALSE	FALSE	Unique entry
TRUE	FALSE	TRUE	FALSE	Address above and below match but names are different.
FALSE	TRUE	FALSE	TRUE	Address above and below are different but names above and below match.

24. We want to use the resulting matrix as a tool to expedite the process for catching anomalies in the data while also transferring the assignment of a unique identifier in the CombMatch column. As we saw above in step 32, however, sometimes the owner name is accurate but there are variations in the address. Since we have used the unique codes for addresses to assign codes for unique owners, we want to check for instances in which names match but addresses do not. We also need to check for entries with a General Delivery address (Note that where "General Delivery" addresses are recorded varies based on year—for 2005 data this information can be found in the "MAIL_ADDR" field and for 2019 data, this can be found in the "OwnerAdd_2" field). This can be accomplished by the following nested IF function formula and placing it in a new column titled "Check":

=IF(OR(AND(H2=FALSE,I2=FALSE,J2=TRUE,K2=FALSE,E2="GENERAL DELIVERY"),AND(H2=TRUE,I2=FALSE,J2=TRUE,K2=FALSE,E2="GENERAL DELIVERY"),AND(H2=TRUE,I2=FALSE,J2=FALSE,K2=FALSE,E2="GENERAL DELIVERY"),AND(H2=TRUE,I2=TRUE,J2=FALSE,K2=TRUE),AND(H2=FALSE,I2=TRUE,J2=TRUE,K2=TRUE)),"CHECK",A2)

The formula above checks for the following combinations:

AddrUMatch	OwnCodeUMatch	AddrDMatch	OwnCodeDMatch	Description
FALSE	FALSE	TRUE	FALSE	No match except address below. “General Delivery” addresses only.
TRUE	FALSE	TRUE	FALSE	Matching addresses above and below but not names. “General Delivery” addresses only.
TRUE	FALSE	FALSE	FALSE	Matching address above only. “General Delivery” addresses only.
TRUE	TRUE	FALSE	TRUE	Only address below not matching.
FALSE	TRUE	TRUE	TRUE	Only address above not matching.

25. For Beaverhead County, the above formula results in 50 “CHECK” values out of 14884 values. We need to review the resulting “CHECK” entries while also preserving the OwnAddrCheck sheet. To do this we will copy and paste the resulting “Check” field into a new sheet called “Check”. Additionally, copy and paste the following fields into the “Check” sheet: the OWNR_NAM1, OWNR_NAM2, MAIL_ADDR, and MAILCITY_S.

26. The process for resolving CHECK entries parallels the process for resolving FALSE entries in step 32 above.

27. Once step 38 is complete, create a new sheet called “Data Check”. Copy and paste the data from Check into this sheet. Format the data as a table and sort OWNR_NAM1 from A to Z. Select all cells with the exception of the headers and use the Remove Duplicates tool under the Data tab to remove all duplicate entries (make sure the criteria for duplication involves all columns in the table, A through E). The results should look similar to the image below. Notice that despite the variations in the data entry for ADAMS CAROLYN and ADAMS FRANK, they are assigned the same unique code (this only refers to the Adams Frank that can be ascertained to be the same owner as ADAMS CAROLYN). These variations were caught in the CHECK steps above. Also notice the ADAMS entries. Presumably, at least two of those ADAMS are the same person and thus the codes should be consolidated but there’s no way to tell for sure with the given information. The accuracy of this entry can only be improved further through ground-truthing. This is an important limitation of this overall method since it depends on inference from

common names and common addresses. When common names do not have common addresses, it's not possible to know how to correctly identify the ownership.

	Check	OWNR_NAM1	OWNR_NAM2	MAIL_ADDR	MAILCITY_S
2	0				
3	2029	2 S PROPERTIES		2337 WAHUPA	HUSON MT598460000
4	0025	3 RIVERS PCS		PO BOX 429	FAIRFIELD MT594360429
5	0025	3 RIVERS TELEPHONE COOP INC		PO BOX 429	FAIRFIELD MT594360429
6	0024	A & B ASSOCIATES		PO BOX 271	LIMA MT597390271
7	1290	AANES JOSEPH M & MARIA L E		505 S ARGENTA ST	DILLON MT597252306
8	0570	ABERNATHY WILLA M		2908 BAKER ST	SAN FRANCISCO CA941233210
9	0433	ABNER EDGAR		24 WHISKEY ROW	LIMA MT597399703
10	0302	ACES INC		4 N WASHINGTON ST	DILLON MT597252514
11	2353	ACHTEN WINONA		328 S CALIFORNIA ST	DILLON MT597252822
12	0957	ACKERSON WADE & TARA		195 AURORA LN	DILLON MT597257355
13	1124	ADAMS CAROLYN		1905 ADAMS LN	DILLON MT597259795
14	1090	ADAMS CAROLYN		550 ADAMS LN	DILLON MT597259734
15	1116	ADAMS FRANK		1710B E PACIFIC AVE	TULARE CA932741107
16	1124	ADAMS FRANK		1905 ADAMS LN	DILLON MT597259795
17	1125	ADAMS FRANK		525 BARNETT AVE	DILLON MT597252901
18	1091	ADAMS FRANK		540 ADAMS LN	DILLON MT597259734
19	1115	ADAMS FRANK		5825 MT HIGHWAY 91 S	DILLON MT597259637
20	1109	ADAMS FRANK		905 T S RANCH RD	BATTLE MTN NV898209501
21	1110	ADAMS FRANK		PO BOX 1308	BELGRADE MT597141308
22	1093	ADAMS FRANK		PO BOX 1405	DILLON MT597251405
23	1105	ADAMS FRANK		PO BOX 749	DILLON MT597250749
24	1126	ADAMS JASON		45 REDROCK DR	BILLINGS MT591026755
25	0759	ADAMS STEVEN SCOTT		1805 LAKEFRONT DR	DILLON MT597259736
26	1137	AFFORDABLE HOMES,INC		2827 CANTEGRA GLN	ESCONDIDO CA920257348
27	3240	AG PRODUCTS COMPANY		1901 BUTTERFIELD RD STE 330	DOWNERS GROVE IL605155448
28	2476	AGBALOG JOSEPH & LILLIAN		PO BOX 41992	EUGENE OR974040556
29	3165	AG-PRODUCTS COMPANY		1901 BUTTERFIELD RD	DOWNERS GROVE IL605150000
30	3377	AGUIRRE PHILIP G JR		1827 WILSON AVE	BUTTE MT597015649
31	0249	ALASKA BASIN GRAZING ASSOC		PO BOX 1046	DILLON MT597251046
32	2591	ALBERI DAVID J & CANDACE H		1540 E GLENDALE ST	DILLON MT597253640
33	0892	ALBERTSON GENNY		658 W 5 S	BLACKFOOT ID832216147
34	1856	ALEBA JOSEPH J		5001 SWEETWATER RD # 7	DILLON MT597256512
35	0653	ALEXANDER M DONALD JR		1 RICHLAND MEDICAL PARK STE 42	COLUMBIA SC292036863
36	0652	ALEXANDER M DONALD JR	REDDING PASCHAL F	1815 SENATE ST	COLUMBIA SC292013818

28. We have yet another data entry anomaly that is resulting in errors related to consolidation. In the MAIL_ADDR column, some include instructions on who to mail to instead of the address. Since neither the owner name or owner address is matchable for such entries these anomalies are the focus of this step in checking the data. For the 2019 Beaverhead County dataset, we are dealing with over 4300 records still. To expedite the process of checking the data, we can perform a similar process to that described in 32-36. Create a new sheet titled "Data Check 2". Copy and paste the values of the "Check" and "OWNR_NAM1" columns of the "DataCheck" sheet into Data Check 2. Create a duplicate column for OWNR_NAM1 and label the first column "Original" because the next step will modify the copied layer. We will need this column to match the resulting codes with the OwnerName and corresponding PARCELID from the Mod1 table.

29. Select all of the OwnerName column in the Data Check 2 sheet and use the Text to Columns tool in the Data tab. Select the "Delimited" option in the wizard and select "Spaces" as the delimiter. This will break the owner names down into separate columns for each name separated by a space in the OwnerName column. For the 2019 Beaverhead County dataset, there are 7 additional columns created from this process. Format the resulting data as a table. The tool should recognize the new columns and include them in the table.

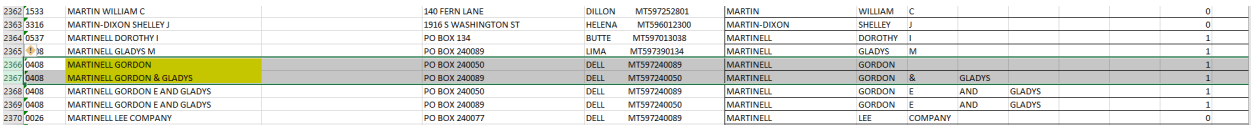
Date: 3/4/2020

30. Following the last column consisting of a component of an owner name, create the following columns and respective formulas (make sure to pay attention to whether or not your data has headers):

Column Header	Formula	Description
MatchU1	=IF(B2=B1,1,0)	Checks if the first component of the owner name matches the first component above. In most cases, this will operate on the last name of the owner or the first word in the business name.
MatchD1	=IF(B2=B3,1,0)	Checks if the first component of the owner name matches the first component below. In most cases, this will operate on the last name of the owner or the first word in the business name.
MatchU2	=IF(C2=C1,1,0)	Checks if the second component of the owner name matches the second component above. In most cases, this will operate on the first name of the owner or the second word in the business name.
MatchD2	=IF(C2=C3,1,0)	Checks if the second component of the owner name matches the second component below. In most cases, this will operate on the first name of the owner or the second word in the business name.
MatchU3	=IF(D2=D1,1,0)	Checks if the third component of the owner name matches the third component above. In most cases, this will operate on the middle name (or initial) of the owner or the third word in the business name.
MatchD3	=IF(D2=D3,1,0)	Checks if the third component of the owner name matches the third component below. In most cases, this will operate on the middle name (or initial) of the owner or the third word in the business name.
MatchCodeU	=IF(A2=A1,1,0)	Checks if the code in the “Check” column matches the code above.
MatchCodeD	=IF(A2=A3,1,0)	Checks if the code in the “Check” column matches the code below.

MatchCodeUD	=IF(OR(T2=1,U2=1),1,0)	Checks if either the code above or below matches for the given code. This couples pairs of matches from above and below.
MatchComp	=IF(AND(OR(N2=1,O2=1),OR(P2=1,Q2=1)),1,0)	Checks for matches with either the first or second name/word above or below.
DataCheckRef	=IF(AND(V2=0,W2=1),"CHECK",A2)	Identifies cases where names match but codes do not with “CHECK”. Otherwise, the current code is entered.

The DataCheckRef column will assist in identifying the entries that should be reviewed manually for further consolidation based on common names. Copy and paste the rest of the data table in Data Check into Data Check 2. The addresses will be needed for reviewing “CHECK” entries. In the example image below, we have a case where the complete owner name does not match and the addresses do not match but it appears that these two entries should share a unique address code.



2362	1533	MARTIN WILLIAM C	140 FERN LANE	DILLON	MT597252801	MARTIN	WILLIAM	C				0
2363	3316	MARTIN-DIXON SHELLEY J	1916 S WASHINGTON ST	HELENA	MT596012300	MARTIN-DIXON	SHELLEY	J				0
2364	0537	MARTINELL DOROTHY I	PO BOX 134	BUTTE	MT597013038	MARTINELL	DOROTHY	I				1
2365	)8	MARTINELL GLADYS M	PO BOX 240089	LIMA	MT597390134	MARTINELL	GLADYS	M				1
2366	0408	MARTINELL GORDON	PO BOX 240050	DELL	MT597240089	MARTINELL	GORDON					1
2367	0408	MARTINELL GORDON & GLADYS	PO BOX 240089	DELL	MT597240050	MARTINELL	GORDON	&	GLADYS			1
2368	0408	MARTINELL GORDON E AND GLADYS	PO BOX 240050	DELL	MT597240089	MARTINELL	GORDON	E	AND	GLADYS		1
2369	0408	MARTINELL GORDON E AND GLADYS	PO BOX 240089	DELL	MT597240050	MARTINELL	GORDON	E	AND	GLADYS		1
2370	0026	MARTINELL LEE COMPANY	PO BOX 240077	DELL	MT597240089	MARTINELL	LEE	COMPANY				0

31. In these situations, preference was given towards *inclusion*. This suggests that errors in this and other datasets that adhere to this methodology will tend towards errors of omission, not commission. Possible consequences of this are the misclassification of certain parcels to a joint address code when they should have been assigned a unique address code. See the data quality control and assurance section at the end of this document for more information. Back to our Data Check 2 process, it clear that many entries that have unique address codes can be consolidated with high confidence using this method. For the 2019 Beaverhead County dataset this identified 346 CHECK cells out of 4338.

32. Manually review the resulting CHECK entries and resolve them in the “Check” column. As this is done, the CHECK entry will be converted to the code. When all desired changes are made, the remaining CHECK values will be for reference during or after ground-truthing. After two iterations of the cleanup process, this number was reduced down to 100 check cells for the Beaverhead 2019 parcel dataset.

33. Now we need to return to the Beaverhead County parcels that had no OWNR_NAM1, OWNR_NAM2, MAIL_ADDR, or MAILCITY_S values. We can loop these back in to the data based on the "OWNCLASS" column mentioned previously.

34. Make a sheet titled "DataCheck3". Let's look back to our original dataset to determine if there are alternate supporting fields that we can use to determine the owner of the parcels with missing data. In this case, it appears that our only two field with relevant information are the

"OWNCLASS" and "GENDESCRIP". We won't need the "GENDESCRIP" field as it does not offer identifying information beyond what is in the "OWNCLASS" field.

35. Copy and paste the values in the OWNCLASS and OWNR_NAM1, fields from our Mod1 sheet to DataCheck3.

36. Format the data as a table, and filter the "OWNR_NAM1" field to select only "Blanks" (it is at the bottom of the value checklist). This leaves us with a list of parcels that were excluded from our homogenization based on owner names and addresses. Now we need to copy and paste the values of our sorted "OWNCLASS" field into column D of DataCheck3. Turn off the data filter applied to our original table, and delete columns A and B, leaving only our new "OWNCLASS" field which contains the parcels we want to compare to our unique address codes. For the Beaverhead 2019 parcel dataset, this leaves 493 parcels that were left out of our methodology.

37. Before we compare these "OWNCLASS" values to our owner names and unique address codes, let's consider the values. Unfortunately, we will have to attach a generic identifier to the private parcels, as these have no additional identifying information. However, we pull own codes for our public parcels.

38. Now let's use a VLOOKUP to match our resulting "OWNCLASS" values with our "OWN_NAM1" values from our DataCheck2 sheet (as we do not have a high number of categories, this could be done manually if desired). Before we do that, we will need to assign a unique identifier to those parcels that do not have ownership information and will not pull a code from our DataCheck2 sheet. In the case of Beaverhead County 2019 parcel data, the "OWNCLASS" values needing an identifier are: "Local Government", "State Government", "US Government" and "Private" (these parcels have no other identifying information to tie them to a existing own code). Navigate to the OwnAddr sheet, and add the new codes to the bottom of the "AddrID"column (keeping in mind that the numbering needs to be continued from the last number in the sequence of codes assigned to "FALSE" values in our Check sheet). The result should look like this:

3443	PO BOX 16, ANACONDA MT597110016	3442		
3444	2989 MARTINEZ WAY, RIVERTON UT840656825	3443		
3445	PO BOX 320006, GLEN MT597320006	3444		
3446	20928 S LAMBERTON AVE, CARSON CA908101	3445		
3447	PO BOX 192, DRUMMOND MT598320192	3446		
3448	PO BOX 98, MENDHAM NJ079450098	3447		
3449	502 CLEAR SPRING DR, HOUSTON TX770797406	3448		
3450		3449	JONES SHAWN L & JANET L	
3451		3450	KOSLOSKY JEFFREY J & MARIA T	
3452		3451	MCDONALDS CORPORATION	
3453		3452	TNMT PROPERTIES LLC	
3454		3453	SMITH JACK R & MARION K	
3455		3454	PENCE DAN T & LOIS M	
3456		3455	ELSE IOLA V	
3457		3456	FLETCHER A BARROWS	
3458		3457	LANGNESS WARREN & LINDA	
3459		3458	HIRSCHY FREDERICK & LYNN HARRI	
3460		3459	NICKISCH RICHARD & CATHY	
3461		3460	PETERSON HAROLD & JOHN	
3462		3461	THOMPSON C DONALD	
3463		3462	Local Government	
3464		3463	State Government	
3465		3464	US Government	
3466		3465	Private	

Notice how previous owner names that were assigned a unique code were recorded for reference.

39. Now we need to add the same "OWNCLASS" values and associated codes to our DataCheck2 "Original" column. Navigate to the DataCheck2 sheet and add the values.

40. Going back to our DataCheck3 sheet, we now can retrieve unique codes for our parcels that lack owner or address information. To get the codes, create a new column heading in column B and title it "AddrID". Now format columns A and B as a table if they are not already.

41. Copy and paste the "Original" and "Check" columns from the DataCheck2 sheet into columns D and E in our DataCheck3 sheet (leaving C as a blank column). Because VLOOKUPS only look right, we need to ensure that "Original" is on the left (in cell D) and "Check" is on the right (in cell E).

42. Now we are ready for the VLOOKUP function to retrieve values for our "AddrID" column. It looks like this: =VLOOKUP(A2, D:E, 2, FALSE). The result is we now have our unique address codes for our problem parcels. To lock in the values, copy and paste values in the "AddrID" back into column B and then delete the "Original" and "check"columns. While these codes could theoretically now be incorporated into a similar check system as we have previously used, there is no need here because of the limited number of categories. For Beaverhead 2019 parcel data, all the parcels lacking ownership data now have a unique address code. The result should look like this:

	OWNCLASS	AddrCode
2	USDI Fish and Wildlife Service	0702
3	Private	3465
4	Local Government	3462
5	Private	3465
6	Private	3465
7	Private	3465
8	USDA Forest Service	0402
9	USDA Forest Service	0402
10	Private	3465
11	Private	3465
12	USDI Bureau of Land Management	0720
13	Private	3465
14	USDI Bureau of Land Management	0720
15	Private	3465
16	USDI Bureau of Land Management	0720
17	State Government	3463
18	Private	3465
19	USDI Bureau of Land Management	0720
20	Private	3465
21	USDA Forest Service	0402
22	Private	3465
23	USDA Forest Service	0402
24	USDI Bureau of Land Management	0720
25	Private	3465
26	Private	3465

43. As a last step, copy and paste the values from our DataCheck3 sheet into our DataCheck2 sheet. Paste the "OWNCLASS" values in our "Original" column (we are using these values as a substitute for owner names), and paste the "AddrID" values into our "Check" column.

44. Create a new sheet and title it “Result”. Copy and paste the “Check” and “Original” columns from the Data Check 2 sheet into the Result sheet. Make sure the Original column is the left-most column (remember that VLOOKUP only looks right). Format it as a table. Now go to the Mod1 sheet and sort A to Z on the OWNR_NAM1 column. Copy the resulting PARCELID, OWNCLASS, and OWNR_NAM1 columns into columns D and E in the Result sheet (column C should be empty), making sure to copy the "OWNCLASS" field for those parcels in Mod1 that have no value for "OWN_NAM1". Use the VLOOKUP function to assign the codes from Check to the respective OwnerName and PARCELID entry. Name that column “OwnCode” as this will be the source for creating a join table to input into the GIS. Review the results for errors and fix the errors if there are any. The resulting table can be used to join tables in QGIS and then perform a dissolve on the OwnCode. The sheet should look something like this:

Original	Check	PARCELID	OwnerName	OwnCode
21 CONSTRUCTION INC	0879	11456332401010000	BUTTERWECK JOSEPH SEBRING	0679
2175 STRATER ROAD LLC	0177	11445803101020000	USDI BUREAU OF LAND MANAGEMENT	9992
A T SERVICE INC	1262	11456534303010000	USDI BUREAU OF LAND MANAGEMENT	9992
ABLE CLINTON SR	1585	11456533404010000	USDI BUREAU OF LAND MANAGEMENT	9992
ABRAHAMSON LYDIA	1217	11445804101010000	USDI BUREAU OF LAND MANAGEMENT	9992
ABRAHAMSON VIRGIL	0921	11456533303010000	STATE OF MONTANA	9993
ABRAHAMSON WARREN & LORI	1083	11423909201030000	MACKEY PENELOPE M	0930
ABRAHAMSON WARREN	1217	11423909201020000	WATSON A HEART LLC	1070
ABRAHAMSON WILLARD & JANICE HOLZHEY	0157	11423702301010000	SALVESON GREG & RUTH	0419
ABRAHAMSON WILLARD & JANICE	0157	11423908101020000	WATSON A HEART LLC	1070
ABRAHAMSON WILLARD & MARILYN	1823	11423908102010000	KNUDSEN LAND & LIVESTOCK LLC	1458
ABRAHAMSON WILLARD E & JANICE R	0157	11423908201010000	LUMSDEN PETER L & CAROL L	0338
ABRAHAMSON WILLIAM	1823	11423606301010000	USDI BUREAU OF RECLAMATION	9996
ABS MT-O LLC	0519	11423907101010000	BARNARD JOHN J & JEANNE	0968
ADAM CONRAD W & LINDA L	0054	11423907201010000	WEBB WILLIAM R & PATTI K &	1256
ADAMS DAVID L & PAUL	0720	11423812201010000	HOLMAN MARK & RONNA L	0834
ADAMS DAVID L	0720	11423705401010000	DOUCETTE RANCH LIMITED PARTNERSHIP	0151
ADAMS LEE A & KATY M	0987	11423811101010000	HOLMAN MARK & RONNA L	0834
AGECOUTAY MARY ESTATE	0565	11424010203010000	USDI FISH AND WILDLIFE SERVICE	9991
ALBERTS RICHARD D & SHARON L	1642	11423811201020000	USDI BUREAU OF LAND MANAGEMENT	9992
ALBUS AARON C	1366	11423811201010000	CROWDER ALLAN J & RHODA M	1622
ALBUS AARON	1366	11423909401020000	SIMS GLEN R & TERRI L	0738

Creating the owner join table

Date: 3/4/2020

1. Create a new workbook and title it “Cad[YEAR]_[COUNTY]CY_JoinProcessing” substituting the appropriate COUNTY and cadastral dataset YEAR. Copy the Mod1 sheet data from the Cad2017_[COUNTY]CY_DataProcessing workbook into the first sheet and title it “Mod1”. Copy the PARCELID table (consisting of the "PARCELID", "OWNR_NAM1", and "OwnCode" columns) from the “Result” sheet and paste it into a new sheet titled “OwnerCodes”. Format it as a table.

2. Use the OwnerCodes table and the VLOOKUP function to import the owner codes into the Mod1 sheet. Create a new column to the left of PARCELID and title it “OwnerCode”. Verify that all cells are matched to an owner code and copy and paste the data values to lock in the owner code.

3. Next, for 2005 data, we need to officially transfer our "OWNCLASS" values for parcels with no owner name information into the "OWNR_NAM1" column. Sort the “OWN_NAM1” column from A to Z, and copy and paste “OWNCLASS” values for those parcels with no “OWN_NAM1” value. To supplement this information, where possible transfer additional address information. For example, we have parcels owned by the United States Department of Interior Fish and Wildlife Service that do have address information, and some that do not. Use the parcels we do have information for to fill in the address columns.

4. For 2019 data, In Mod1, create a new column at the end of the table and title it “CityState” use a CONCAT function like this “=CONCAT(G2,", ",H2)” to create a combined city and state location for the owner. In the case of 2005 Beaverhead County cadastral data, this is not necessary as a concatenated field already exists "MAILCITY_S".

5. For 2019 data (or other years with an "OwnerZipCo" field) Create a new column to the right of CityState and title it “ZipCode”. Use the LEFT function to pull the first five characters from the OwnerZipCo column. This will result in a column of 5-digit zip codes. For 20005 data, use this formula to pull out the zip code from the "MAILCITY_S" field: =VALUE(RIGHT(G2,LEN(G2) - MIN(SEARCH({0,1,2,3,4,5,6,7,8,9}, G2&"0123456789")) +1)) where G2 is the first cell that contains the zip. Then use a LEFT function, which looks like =LEFT(H2, 5) where H2 is the new 9 digit zip code that was extracted. This results in a 5 digit zip code which can be copied/pasted to lock in values.

6. Use the city and state provided to match the place with its respective county using census data. To do this, access HUD USPS Zip Code Crosswalk files here. After the Zip Code Crosswalk Files have been downloaded, run a VLOOKUP to extract the "County" column values within the Crosswalk file. This code includes the state and county FIPS codes in a five digit number. For 2005 data, we now have the city, state, zip code, and FIPS code but no county name. To determine the county name from our already calculated FIPS code, we need to download a FIPS code dataset from the U.S. Census Bureau here (search for “FIPS County” and download the most current year). It should be noted here that it is necessary to tease out the County name from the name + “county”. For example, in the Census dataset, I used this function to trim the last word of the county/city column: =LEFT(TRIM(C1),FIND("~",SUBSTITUTE(C1," ","~",LEN(TRIM(C1))-LEN(SUBSTITUTE(TRIM(C1)," ",""))))-1). It is also necessary to combine the split state and county FIPS code columns into one column The last step is to copy and paste the combined FIPS code and adjacent county name column into our Mod1 sheet and use a

VLOOKUP function to assign county names to our parcels (first a column titled “county” must be added).

7. As one last database normalization step, let’s input “No data” values for our parcels lacking ownership and address information. To do this, sort the “MAIL_ADDR” column A to Z, and input “No data” into all the entries with no address information (do this for every column that we won’t have info for, including address, city, state, county, and FIPS).

8. In a new sheet, copy and paste the resulting Mod1 dataset. Title the sheet “Reduction”.

9. Remove the following columns from the Reduction dataset: PARCELID, "OWNR_NAM1" "OWNCLASS", "OWNR_NAM2", "MAIL_ADDR", "City", "State", "Zip" OwnerAdd_2, The only columns that should remain are OwnerCode, OWNR_NAM1, and STCY_FIPS (the name for the column with the state and county FIPS codes for the owner).

10. Use the delete duplicates tool to remove duplicates where the owner code, owner name, and county and state location of the owner are the same. This means using all three columns as criteria.

11. Create a new sheet titled “Join”. Copy and paste the OwnerCode column from Mod1 into the first column of this sheet and remove the duplicates. This produces a field with only unique owner codes. We will use the Reduction table to populate some informative fields in this table.

12. Create two new columns in the Join sheet titled “Names” and “Parcels”. Use the COUNTIF function to count the number of unique names and parcels associated with the unique owner code. Note that you will have to use the Mod1 sheet in order to count the number of unique parcels associated with each owner code (due to our Reduction sheet not containing the “ParcelID” field. Make sure to copy and paste the values after the function has run to lock in the values.

13. Use the VLOOKUP function to add additional information about the owner (name, address, state, place, FIPS) into the table in the Join sheet. Note that for owner codes where there is more than one owner name, the VLOOKUP function will only use the first entry. The quantity in Names provides a reference for the reader interested in seeing all the names associated with an owner code. The Reduction sheet provides a reference table as does the Mod1 table.

14. Once the additional fields have been populated and checked in the Join sheet, copy and paste the resulting dataset to a new workbook, title it “Cad[YEAR]_[COUNTY]CY_Join”. Make sure the OwnerCode is the left-most column in the table. Save an excel workbook version of this data and a CSV version. The CSV version is needed for import into the GIS.

15. To complete the process of producing a consolidated ownership vector file in the GIS, perform a join using the PARCELID and OwnerCodes from the Mod1 table from the “Cad[YEAR]_[COUNTY]CY_JoinProcessing” spreadsheet. The PARCELID field will be the join field. Save a copy of the resulting vector file. In the copy version, delete all fields except the OwnerCode field. Perform a dissolve on the copy using the OwnerCode field. This results in the consolidated ownership vector file. Import the “Cad[YEAR]_[COUNTY]CY_Join” csv table and use the OwnerCode field to join the table to the vector file attribute table. Use the geometry

calculator to calculate the area of the resulting consolidated parcels. Save a copy of the resulting vector file. Open the saved copy and delete the redundant OwnerCode field. This completes the process.